

Math Olympiad Trainer Academy

AY 25/26 — Cycle I | Exercise Collection

Part I — Geometry

Geometrical figures and all related material are available in Google Classroom.

Exercise 1

For points M and N on sides BC and CD of square $ABCD$, we have $\angle MAN = 45^\circ$. The feet of the altitudes from M and N in triangle AMN are P and Q . Show that B, P, Q are collinear.

Exercise 2

Points A, B, C, X, Y on a circle satisfy the condition $AB = BC$. The intersection of AX and BY is P ; the intersection of BX and CY is Q . Show that $PQ \parallel AC$.

Exercise 3

For points A, B, C, D on a circle centered at O , lines AB and CD intersect perpendicularly at point P . The foot of the perpendicular from P to AO is K , and the circumcircle of KPB intersects PC again at Q . Show that $CP = QD$.

Exercise 4

Circles ω_1 and ω_2 intersect at points A and B . A line through B intersects ω_1 and ω_2 again at P and Q respectively. A line through Q is tangent to ω_1 at T and intersects ω_2 again at X . Show that line AT is an angle bisector of $\angle XAP$.

Exercise 5

In acute triangle ABC , the feet of the altitudes are A', B', C' . Let P be one of the intersection points of $B'C'$ with the circumcircle. Let BP and line $A'C'$ intersect at Q . Show that $|AQ| = |AP|$.

Exercise 6

In triangle ABC , the reflection of A with respect to BC is P , and the second intersections of PB and PC with the circumcircle ω of ABC are X and Y respectively. Show that the circumcenter of PXY lies on ω .

Exercise 7

In triangle ABC , the orthocenter is H ; the intersections of BH and CH with AC and AB are D and E respectively. Let P be the reflection of E with respect to AH , and Q be the reflection with respect to AC . Show that D, P, Q are collinear.

Exercise 8

In triangle ABC , points X, Y, Z are on BC, AC, AB respectively. The second intersections of the line AX with the circles $(BXZ), (CXY)$ and (AYZ) are M, N , and P respectively. Show that

$$\frac{MP}{NP} = \frac{BX}{CX}.$$

Exercise 9

In triangle ABC , points X, Y, Z are on BC, AC, AB respectively. The second intersection of the circles (BXZ) and (CXY) is P , the second intersection of the circle (BYZ) and the line BC is M , and the second intersection of the circle (CYZ) and the line BC is N . Show that PX, MY, NZ are concurrent.

Part II — Number Theory

Exercise 10

For any positive integer n , let $M(n)$ denote the greatest divisor of n which is distinct from n . Ten positive integers satisfy

$$a_1 < a_2 < \cdots < a_{10}$$

and

$$M(a_1) > M(a_2) > \cdots > M(a_{10}).$$

Show that $a_{10} > 500$.

Exercise 11

Is there an arithmetic sequence of length 1000 any two terms of which are coprime?

Exercise 12

Given any 10 consecutive integers, show that there's one of them coprime to all others.

Exercise 13

n is a positive integer and d_1 and d_2 are divisors of n . If

$$\gcd\left(d_1, \frac{n}{d_2}\right) = \gcd\left(d_2, \frac{n}{d_1}\right),$$

show that $d_1 = d_2$.

Exercise 14

Given distinct positive integers $a_1, a_2, \dots, a_{100}$, denote

$$b_i := a_i + \gcd(a_1, a_2, \dots, \widehat{a_i}, \dots, a_{100}).$$

Said in words, b_i is equal to a_i plus the greatest common divisor of all the other 99 integers. At least how many distinct values appear among $\{b_1, b_2, \dots, b_{100}\}$.

Exercise 15

Positive integers a, b, c, d satisfy $ad = bc$. Show that there exist positive integers p, q, r, s such that

$$a = pq, b = pr, c = qs, d = rs.$$

Exercise 16

Let a, b, c be positive integers such that

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$$

and $\gcd(a, b, c) = 1$. Prove that $a + b$ is a perfect square.

Exercise 17

a, b, r, s are positive integers such that

$$\frac{a}{600} + \frac{b}{700} = \frac{r}{s}$$

and the fractions $\frac{a}{600}$ and $\frac{b}{700}$ are already simplified (which means that the numerator and the denominator have no common factor). What is the least possible value of s ?

Exercise 18

A positive integer consists of 2's and 0's in its decimal expression (for example, 22020022020002). Can it be a perfect power?

Exercise 19

n and N are positive integers with $N > n^{n-1}$. Show that there exist distinct prime numbers $p_1, p_2, \dots, p_n$ satisfying

$$p_1 \mid N+1, p_2 \mid N+2, \dots, p_n \mid N+n.$$